AML EXECUTIVE REPORT

Aurelia Bank Financial Crime & AML Intelligence Control Tower

KYC risk, transaction monitoring, graph analytics, alert triage and human-led case governance

Controlled synthetic portfolio | Cutoff 20 August 2026

Prepared by Murat Miraç Gedik

Portfolio demonstration — not a production or regulatory system

Executive decision

Clear the KYC review backlog without weakening a currently controlled alert and case operation.

49.6%	7.0%	0
KYC reviews overdue	**Customer alert rate**	**Open SLA breaches**
892 customers vs 35% ceiling	126 of 1,800 customers	50 active cases

Decision requested

Approve a risk-ranked KYC remediation program: alerted and High-KYC-risk customers first, separately funded weekly capacity, and monthly review of backlog, SLA and validation evidence.

Why it matters

- Stale KYC context weakens the quality of transaction-monitoring investigations.
- The current alert rate is inside the 12% internal ceiling and active cases remain within SLA.
- Unplanned KYC work could consume the same analysts who protect the zero-breach position.

Control boundary

Alerts are investigative leads. This system cannot submit a ŞİB/STR, restrict a customer or establish criminal conduct.

MASAK Measures Regulation

Portfolio and data posture

The snapshot is broad enough to demonstrate the complete control flow while remaining free of real personal or bank data.

Customers	Accounts	Transactions	TRY-equivalent value
1,800	2,250	48,412	TRY 416.6m

Aurelia Bank Financial Crime & AML Intelligence Control Tower

Data classes

- Controlled synthetic: customer, account, transaction and relationship records.
- Synthetic validation truth: 92 injected subjects, isolated from every detection module.
- Synthetic internal taxonomy: fictitious jurisdiction XH; no official country-risk rating.
- Official public methodology: MASAK, FATF and UN sources used only for governance context.
- Derived analytics: KYC scores, hits, anomalies, graph features, alerts, cases and controls.

Analytical method

Four independent signals converge in an explainable score; one authorised human decision remains.

Layer	Method	Output	Guardrail
KYC	Additive policy score	0–100 inherent risk	Reason components exported
Rules	Six deterministic scenarios	Evidence-bearing hit	Internal parameters, not legal thresholds
Behaviour	Isolation Forest	Percentile anomaly	Cannot create a case alone
Graph	NetworkX	Degree, PageRank, component, cycles	Centrality is not guilt
Triage	Weighted linear score	High / Medium / Low	Every case requires human review

Alert score

60% scenario severity + 16% KYC + 14% behavioural anomaly + 10% graph evidence. High priority begins at 80 and Medium at 62.

Validation separation

The detection API accepts no truth table. Precision, recall and F1 are calculated only after alerts exist, at scenario/customer grain.

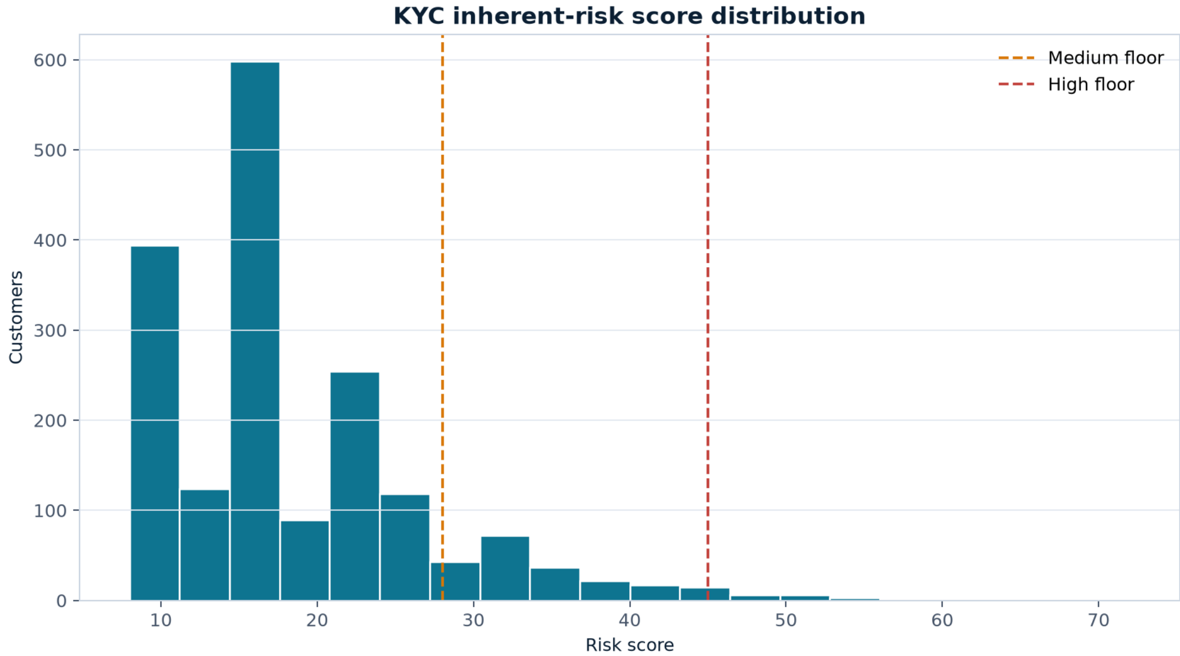
Amount-independent assessment

MASAK's framework treats suspicious transaction assessment as independent of amount. Every numeric value in this project is therefore labelled as an internal monitoring parameter.

FATF risk-based approach for banking

KYC risk and review backlog

Only 29 customers are High inherent risk; the control issue is the breadth of overdue periodic reviews.



Band	Customers	Share	Review implication
Low	1,577	87.6%	Standard risk-ranked review
Medium	194	10.8%	Accelerated where alerted or overdue
High	29	1.6%	First remediation cohort

Recommended clearance order

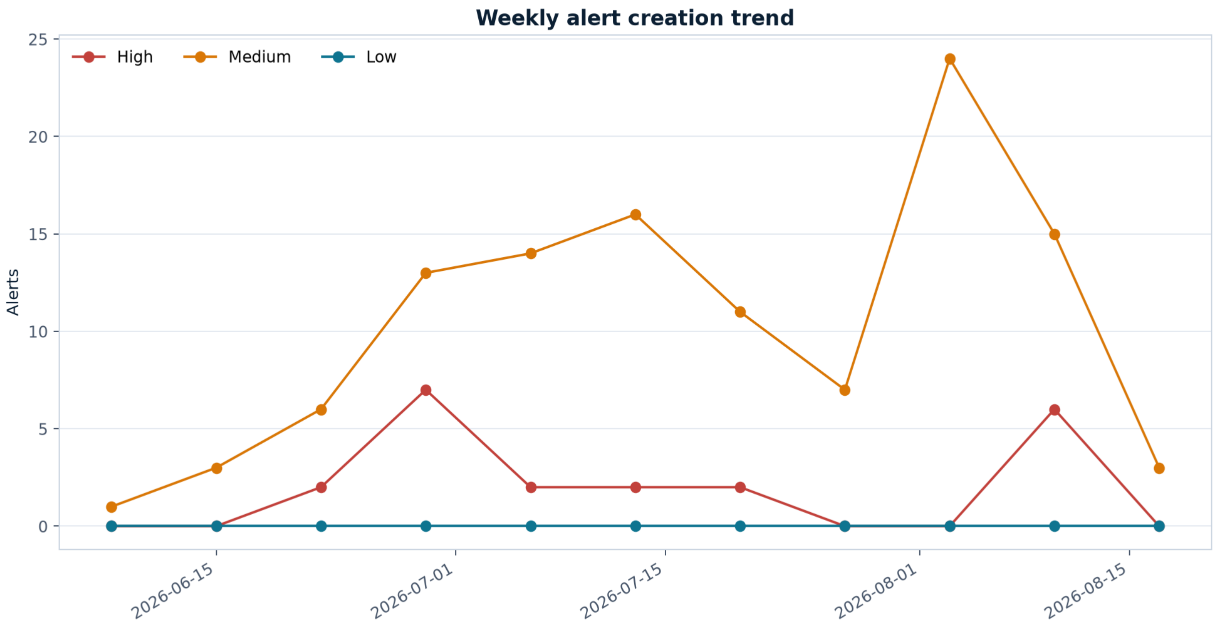
- Alerted and High-risk customers.
- Remaining alerted customers with overdue review dates.
- Non-alerted backlog ranked by KYC score and control gaps.
- Weekly burn-down with separate capacity and quality sampling.

[FATF beneficial-ownership guidance]

Transaction-monitoring scenarios

Six explainable scenarios produce 134 alerts across 126 customers—a 7.0% customer alert rate.

Scenario	Alerts	Internal evidence focus
Circular transfer	30	Three-node high-value cycle
Rapid movement	23	Material inflow and outbound ratio
Structuring	23	Repeated cash inflows below internal parameter
High-risk geography	21	Fictitious XH value and count
Dormant reactivation	20	Dormancy flag and material activity
Funnel / mule	17	Counterparty fan-in and concentrated outflow



Evidence standard

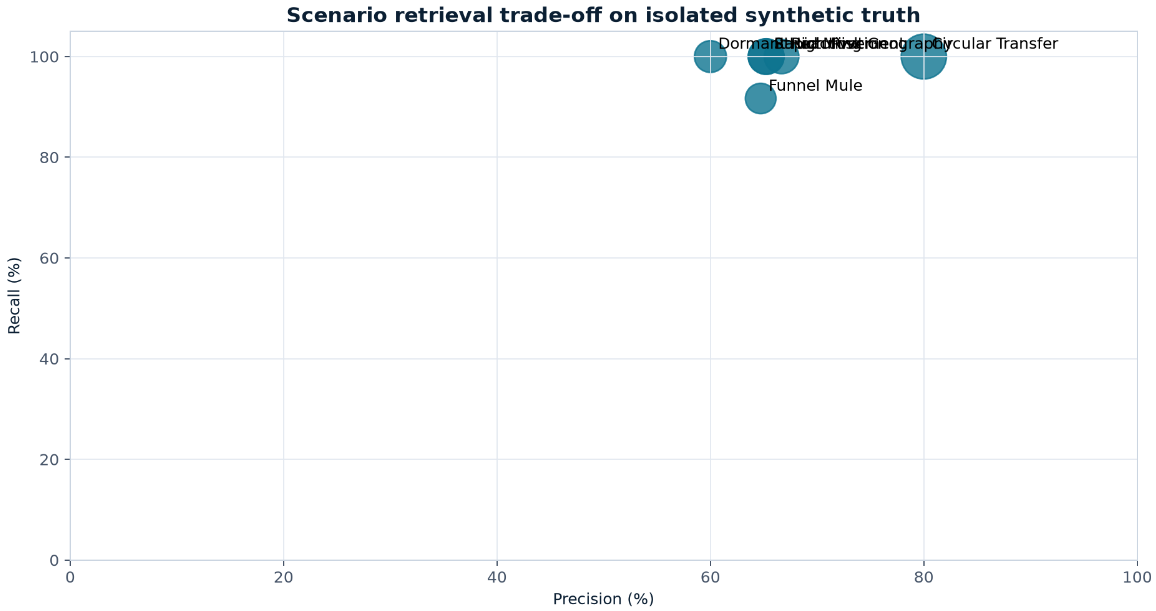
Every hit keeps the customer, account, time window, source transaction identifiers, amount, observed value, internal parameter and plain-language rationale. Legitimate explanations must be tested before escalation.

[MASAK sectoral suspicious transaction guides]

Synthetic validation

Deliberate lookalikes create false-positive pressure and prevent a misleading perfect-precision result.

Scenario	TP	FP	FN	Precision	Recall
Circular Transfer	24	6	0	80.0%	100.0%
Dormant Reactivation	12	8	0	60.0%	100.0%
Funnel Mule	11	6	1	64.7%	91.7%
High Risk Geography	14	7	0	66.7%	100.0%
Rapid Movement	15	8	0	65.2%	100.0%
Structuring	15	8	0	65.2%	100.0%

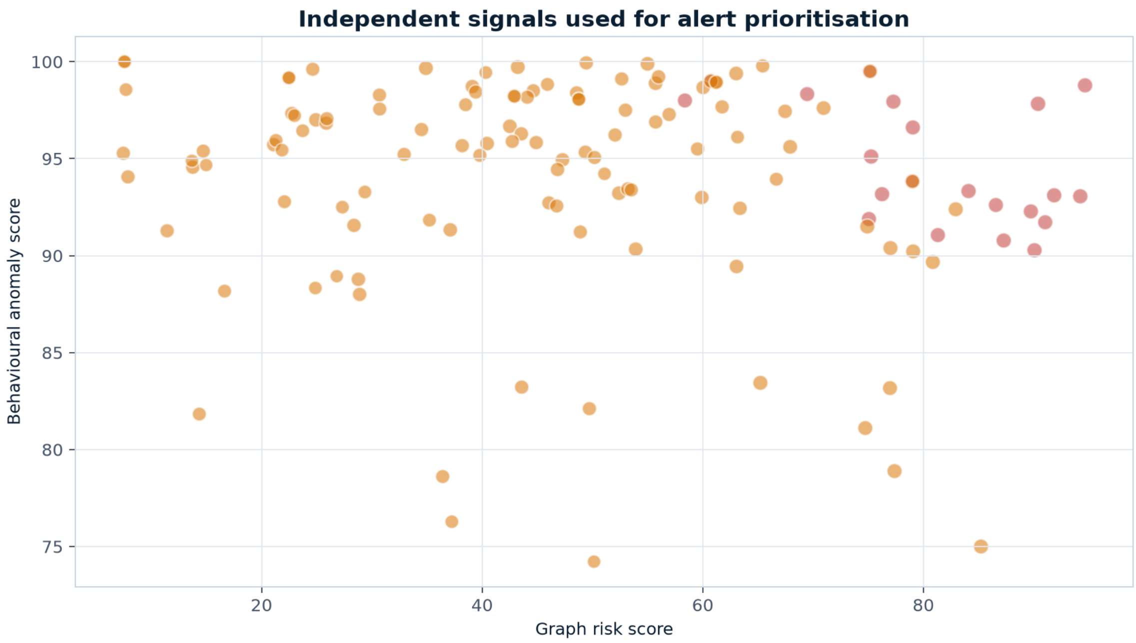


Overall result

Precision 67.91%, recall 98.91% and F1 80.53%. Funnel/Mule retains one false negative. These results describe controlled synthetic retrieval only and cannot predict production effectiveness.

Behavioural and graph intelligence

Independent signals improve ranking while deterministic rule severity remains dominant.



Signal	Exported evidence	Weight
Isolation Forest	Count, value, counterparties, max transaction, turnover deviation, cash, geography and out/in ratios	14%
NetworkX	In/out degree, weighted PageRank, component size and short-cycle flag	10%

Interpretation guardrail

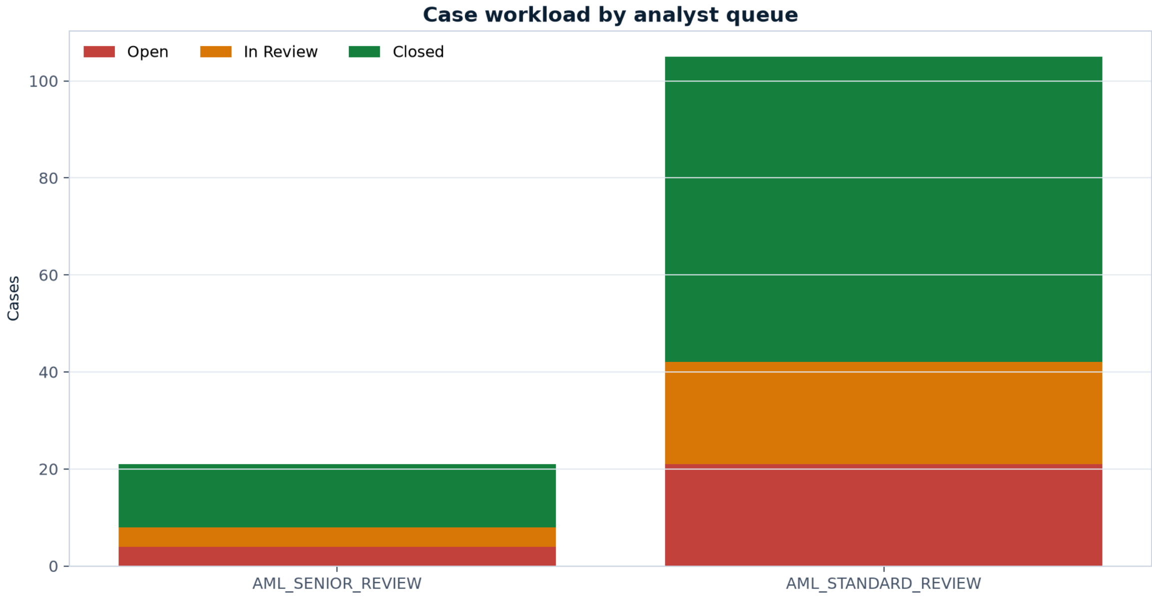
A central customer, connected component or high anomaly score may have a legitimate business explanation. Analysts must record both supporting and contradictory evidence; neither signal can create a case independently.

Graph snapshot

30 customers appear in short high-value cycles. The broad internal-transfer network forms one connected component, so component membership itself carries little discriminatory value.

Case operations and human oversight

The current queue is within the five-day capacity proxy and has no active SLA breaches.



Status	Cases	Control implication
Closed	76	Disposition retained; no automated regulatory outcome
In review	25	Assigned analyst queue
Open	25	Within current SLA
Open SLA breach	0	Zero-tolerance target met

Required investigator sequence

- Verify evidence lineage and transaction chronology.
- Compare activity with expected customer purpose and current KYC.
- Review beneficial ownership, source of funds and counterparty context.
- Document legitimate explanations and contradictory evidence.
- Leave the ŞİB/STR decision to the authorised compliance/MLRO function.

90-day action plan and source register

Reduce backlog first, tune scenarios second, and make production-readiness conditional on independent controls.

Window	Action	Evidence of completion
0–30 days	Clear alerted + High-risk KYC cohort; protect analyst capacity	Weekly backlog, SLA and QA report
31–60 days	Segment scenario parameters; sample outcomes; challenge bias and drift	Approved tuning and validation record
61–90 days	Design live screening, privacy, access, audit and case integration	Security, Legal, Privacy, MLRO and Model Risk sign-offs

Primary official references

MASAK sectoral ŞİB guides
MASAK guide update notice — 11 September 2025
Law No. 5549
FATF Recommendations — updated June 2026
FATF Recommendation 16 update — June 2025
UN Security Council Consolidated List

Final decision

Approve the remediation sequence, assign accountable KYC and AML Operations owners, and require monthly evidence review. This report is a portfolio demonstration—not regulatory certification, legal advice or a production control.